Research Evolution | DUOL

Baseline and provenance

Edition A below reproduces the existing source without revising its content. docs/edition-a.md remains byte-identical to the original repository draft. PDF formatting and pagination are new; the underlying text is unchanged. I did not submit Edition A to Brightspace before AI-assisted work began. Preserving the draft does not establish a checkpoint receipt; checkpoint eligibility and recovery are handled with the instructor through the course process.

Edition A - unchanged

Project 1 - Edition A: Duolingo, Inc. (NASDAQ: DUOL)

Decision and user

This memo supports an investment committee deciding whether the fund should initiate a position in Duolingo. The decision is whether the available evidence supports committing capital now, continuing research without committing capital, or rejecting the opportunity.

Company and valuation date

- Company: Duolingo, Inc. (NASDAQ: DUOL; CIK 0001562088)

- Valuation date: September 1, 2026

- Primary source reviewed: Duolingo's Form 10-K for the fiscal year ended December 31, 2025, accession 0001628280-26-012494, filed February 27, 2026 (local filing; SEC filing detail)

Initial thesis: watch / defer

Do not initiate yet. Duolingo shows attractive product-led growth, strong engagement, rising paid conversion, positive operating cash flow, and a debt-free balance sheet. Those facts justify continued research. They do not yet justify buying the stock because this Edition A has not established the market price, enterprise value, expectations embedded in that price, performance after December 2025, or whether growth can remain efficient as the business scales.

The investable thesis to test is that Duolingo's large and increasingly engaged free-user base creates a durable, low-acquisition-cost funnel into paid subscriptions. If paid conversion and retention compound without sales and marketing growing faster than revenue, revenue and cash flow may scale faster than the cost base. The counter-thesis is that engagement or subscriber growth normalizes while valuation continues to require exceptional growth, producing poor returns even if the company remains a good business.

Known evidence

Unless otherwise noted, the evidence below comes from the 2025 Form 10-K cited above.

- Scale and engagement: for the three months ended December 31, 2025, average monthly active users were 133.1 million, average daily active users were 52.7 million, and DAU/MAU was 39.6%. DAUs grew 30% year over year, faster than MAUs at 14%, indicating increased usage frequency.

- Monetization: paid subscribers reached 12.2 million at year-end, up 28% from 9.5 million. Approximately 9% of MAUs were paid subscribers. Subscription bookings rose 36% to $996.3 million, and total bookings rose 33% to $1.158 billion.

- Financial growth: 2025 revenue was $1.038 billion, versus $748.0 million in 2024, an increase of approximately 39%. Operating income was $135.6 million, compared with $62.6 million.

- Cash generation: operating cash flow was $387.8 million and reported free cash flow was $360.4 million. Stock-based compensation was $137.4 million, so cash flow should not be treated as fully equivalent to owner earnings without considering dilution.

- Balance sheet: cash and cash equivalents were $1.036 billion, with another $239.2 million of short- and long-term investments. No borrowing debt was reported. Operating-lease liabilities totaled approximately $101.0 million and are analyzed separately from financial debt.

- Share evidence: the 2025 diluted weighted-average share count was 48.308 million, versus 45.773 million basic weighted-average shares. Year-end Class A and Class B shares outstanding totaled 46.628 million. The diluted count included founder awards, stock options, and restricted stock units.

- Earnings-quality caution: 2025 net income of $414.1 million included a one-time $256.7 million income-tax benefit from releasing a deferred-tax valuation allowance. Reported net income therefore overstates recurring operating profitability.

- Distribution concentration: 62% of revenue came through the Apple App Store and 20% through Google Play. Duolingo generally paid those platforms 15%-30% of payments processed through their in-app systems, creating platform-policy and take-rate risk.

- Management's metric definitions matter: MAUs and DAUs are period averages, while paid subscribers are measured at period end. Comparisons must preserve those definitions rather than treating all three as point-in-time counts.

Consequential assumptions

- Product-led acquisition remains durable: user growth and engagement continue to arise primarily from product quality, retention, and word of mouth rather than increasingly expensive paid marketing.

- Engagement converts into economics: higher DAU/MAU ultimately supports paid conversion, retention, advertising inventory, or pricing power rather than engagement that is costly but weakly monetized.

- Subscription growth remains profitable after app-store fees, content and AI infrastructure costs, stock-based compensation, and dilution.

- Reported user and subscriber metrics are consistently measured and are sufficiently resistant to duplicate accounts, changing definitions, and product-mix effects.

- The cash balance is available to shareholders or productive reinvestment and is not consumed by acquisitions or spending that earns below the fund's required return.

- A suitable purchase price must provide an adequate return under a material deceleration scenario; business quality alone is not enough.

Unknowns

- DUOL's share price, market capitalization, enterprise value, and valuation multiples on September 1, 2026.

- Results and guidance released after the 2025 year-end, including current user growth, subscriber growth, bookings, margins, cash flow, and dilution.

- Subscriber retention, churn, cohort behavior, family-plan economics, conversion by geography, and the contribution of price increases versus subscriber volume.

- Unit economics, including customer acquisition cost, lifetime value, incremental contribution margin, and the direct cost of serving AI-intensive features such as Duolingo Max.

- How much of recent engagement growth is durable versus driven by temporary product launches, marketing, notifications, or changes in metric measurement.

- Revenue and gross-profit concentration by product, geography, and platform, including the economics of Duolingo Max and the Duolingo English Test.

- Management's capital-allocation priorities for the cash and investment balance.

- The market's current growth and margin expectations and the downside if Duolingo merely performs well rather than exceptionally.

Research plan

1. Establish the valuation-date capital structure using the September 1, 2026 closing price, current basic shares, in-the-money options and awards, cash, investments, debt, and lease liabilities. Reconcile weighted-average diluted shares to period-end fully diluted shares.

2. Read all 2026 10-Qs, earnings releases, call transcripts, and relevant 8-Ks available by the valuation date. Build a quarterly history of MAUs, DAUs, DAU/MAU, paid subscribers, bookings, revenue, gross margin, operating margin, stock-based compensation, and free cash flow using consistent definitions.

3. Decompose growth into user growth, paid conversion, pricing, retention, advertising, and other products. Identify where public disclosure cannot support the decomposition and avoid false precision.

4. Evaluate operating leverage by comparing revenue growth with cost of revenue, research and development, and sales and marketing. Adjust earnings and cash flow for the one-time tax benefit and explicitly account for stock-based compensation and dilution.

5. Assess competitive and platform risks: compare Duolingo's engagement, pricing, learning efficacy, AI features, and distribution economics with credible substitutes; test sensitivity to Apple and Google fees or policy changes.

6. Build base, upside, and downside five-year scenarios for subscribers, average revenue per subscriber, non-subscription revenue, margins, reinvestment, dilution, and terminal valuation. Use reverse valuation to identify what the current price already assumes.

7. Define decision thresholds before recommending initiation: required expected return, acceptable downside, evidence that growth remains product-led, and observable conditions that would invalidate the thesis.

Partner-generated falsification question

What evidence would falsify the thesis if Duolingo's paid-subscriber growth fell below 20% while sales-and-marketing spending rose faster than revenue, indicating that growth increasingly depends on paid acquisition rather than durable product-led conversion?

Edition B - integrated valuation model

Prepared October 17, 2026 local time; valuation remains September 21, 2026. Intended user: a buy-side committee with no existing position. Investment horizon: 3-5 years. Recommendation: **Watch / Defer** at the dated $149.42 price. The corrected five-year pro-forma DCF gives **$137.57**, with conditional low/base/high operating cases of **$114.20 / $137.57 / $165.01**.

The application links the operating forecast, valuation, sensitivity analysis and validation in one model. model.py follows Lab 10's revenue-to-statements-to-cash sequence and Lab 6's FCFF valuation method. analysis.py carries forward Lab 11's reset-and-change-one-input discipline. Lab 08 supplies the peer policy, Lab 09 the independently runnable training answer, and Lab 12 the decision narrative. A small panel, explicit validation and final documents close the project-specific gaps. The previous expanded implementation is preserved outside this submission as an archive; it is not another active engine.

Source	Retained contribution	Final correction or addition
Edition A and September research	Thesis, assumptions, evidence, unknowns	Preserve A; qualify tax-contaminated earnings and unresolved claims in B
Lab 6	WACC, FCFF DCF, terminal share, sensitivity and reverse DCF methods	Drive DCF with the operating pro forma instead of extrapolating reported FCF; reconcile date and shares
Lab 08	Peer policy, GAAP earnings gate, merger exclusion and method disagreement	Add consistent DUOL EV and equity ratios; retain absence of an eligible independent peer target
Lab 09	ABG known-answer and deliberate cash failure	Retain the $291.75 training fixture as validation, not a DUOL FCFF model
Lab 10	DUOL five-year drivers and cash-last accounting	Show full statements; add cash/FCFF identities, H1 actuals, dated liquidity, asset roll-forwards and terminal reinvestment
Lab 11	OAT EBIT/FCFF/value and reset behavior	Fix AI growth versus margin inconsistency; preserve original prediction history; test equal-width rankings
Lab 12	Presentation logic, monitoring and partner questions	Translate reasonably-valued language into a conditional committee action and an explicit final range
Final project requirements	Usable workbench, complete validation and deliverables	Bounded controls, reset, visible failures, executed report, cold run, memo and disclosure

Why the value changed

Lab 6's $251.99 result starts with reported company CFO less capital spending and uses September 10 price evidence, 48.308m weighted-average diluted shares and 8.5286% WACC. Labs 10-12 use $140.18, a September 21 price, 50.7m diluted shares and 8.2764% WACC. The final model produces $137.57 after correcting the later lab's timing, liquidity, cash-flow and terminal conventions. This is a multi-change reconciliation, not an isolated experiment attributing the entire $2.61 difference to a single correction.

The final engine uses FY2025 historical statements and filed H1 2026 balances/flows, forecasts H2 separately, and displays full-year 2026 through 2030. DCF excludes the elapsed portion of H2 at September 21 using an explicitly approximate uniform-accrual convention. The dated equity bridge adds estimated excess liquidity only once, including a $100m operating reserve deduction. Final-year working-capital investment is recomputed at mature terminal growth rather than carrying the high-growth year mechanically into perpetuity. D&A is allocated across PPE/software assets rather than charging all depreciation to PPE while software never amortizes.

The earlier expanded implementation and initial consolidation both produced $136.89. The final cash-bridge correction increases that estimate to $137.57. The bridge now uses accounting cash accumulation rather than economic FCFF, and excludes the $0.675m repurchase payable from operating working capital. It settles in financing without reducing equity twice. This correction is separately tested; it does not change the operating growth assumptions. The model uses Python's standard library, with retained source evidence and original file hashes.

Normalization and company-specific drivers

FY2025 reported net income is $414.065m. Removing the $256.7m valuation-allowance release alone yields $157.365m. The selected analytical earnings measure is instead $137.409m: remove other nonoperating income and replace the entire reported tax line with a normal 24% tax assumption applied to EBIT plus interest income. Operating NOPAT excludes interest. SBC remains an economic operating cost, and capital spending includes both software and PPE. These definitions are reconciled in outputs/normalization.json and the contamination catalog; none is relabeled as GAAP.

The 2026 $1,207m revenue and 71.6% gross-margin anchors are guidance-based. The later 18% to 10% growth path, margin recovery, declining SBC intensity and $100m reserve are judgments. Subscription conversion/retention and bookings support revenue and deferred-revenue reasoning; AI delivery costs affect margin. Neither channel is separately identified from a causal statistical model. Each of the 24 material assumptions has a value, basis, challenge and evidence that would cause revision in the assumption ledger and challenge record.

Lab 11 correction and prediction history

Both current Lab 11 named drivers change the same revenue-growth vector, at different widths. One prose passage incorrectly describes a gross-margin shock. Equal growth shifts necessarily give identical outputs, so the larger observed span does not establish that one business mechanism intrinsically matters more.

Git commit 3c6dab3 preserves an earlier prediction-only record with AI efficiency modeled through gross margin. Later commits e3aefff and f9d2c2e add results and change the AI channel to revenue growth. The final package retains original and current versions under evidence/ and records the mismatch in docs/locked-changed-input-record.md. The old prediction is not silently rewritten. I wrote the original prediction myself in class with AI closed; the prediction-only commit precedes the matching result record, and the original gross-margin results were reproduced from preserved code. The exact historical execution time and a contemporaneous formal action statement are not separately established.

The final OAT analysis separately tests revenue growth, gross margin and SBC, plus the other assumptions. Growth leads under the wider +/-3pp growth range; under equal +/-1pp shocks, margin slightly exceeds SBC for value, and both exceed growth. Both EBIT/FCFF and valuation changes are visible. These are conditional stress comparisons, not probabilities.

Decision, disagreement and monitoring

At the dated market price, the base DCF does not provide a valuation cushion. Terminal value is 84.34% of EV; a 20% terminal ROIC stress reduces value to approximately $126.01. This is material economic uncertainty, even when accounting checks pass. Reverse DCF separately solves approximately 7.78% WACC, 3.53% perpetual growth, or a +2.32pp growth-path shift to match the market price. It does not reveal investors' actual beliefs.

Five named downside stresses each change one disputed judgment: no 2027 reacceleration ($127.16), growth at the FY2026 bookings pace of about 11% ($122.63), beta 1.0 ($125.56), total-return beta ($134.15) and terminal ROIC 20% ($126.01). The 2027 reacceleration to 18% is my weakest assumption because bookings guidance (+10.9% for FY2026, +8.9% for Q3) leads subscription revenue. The regression beta of 0.85 has an R-squared of only 0.045. Every stress lowers value, so none reverses Watch / Defer; together they move the case toward Do Not Initiate.

Market EV/sales is 5.99x under the diluted-equity convention; equity/reported earnings is 18.30x but equity/normalized earnings is 55.13x. Reported-tax effects and differing forecast horizons explain why these measures suggest different valuation burdens. The COUR/UDMY exclusions remain: negative annual GAAP earnings and post-merger entity mismatch prevent a defensible peer target. These incompatible methods are not averaged; own-company multiples do not provide independent peer corroboration.

Watch the next two reports for revenue/bookings, margin, SBC and investment needs. Gross margin below 70.6%, growth 3pp below the path, SBC above 17% of revenue, or FY2027 bookings growth below 13% requires a rebuild; persistent thesis failure supports Do Not Initiate. A proposed initiation rule requires both price at or below the current low-case estimate and evidence supporting the base economics, with updated dated inputs. These are judgment conditions, not guaranteed outcomes.

Attribution and remaining personal evidence

The IBM partner challenged the growth path in Lab 10 and the AI cost/revenue channel and sensitivity ranges in Lab 11. The Apple partner challenged the AI-versus-paid-conversion interpretation in Lab 12. Their exchanges and responses are credited in docs/assumption-challenges.csv. Codex (gpt-6-astra, low reasoning) assisted consolidation, corrections, testing and drafting on October 16-18, 2026. Claude Code (Claude Opus 5.5) reviewed the package against the course requirements on October 18, 2026, adding the beta reliability statistics and named stresses. Dispositions are in the validation record.

Item	Edition A	Edition B	Primary driver of change	Evidence/test
Thesis/recommendation	Watch / defer; no price or valuation	Watch / Defer; $114.20-$165.01 range vs $149.42	Own analysis	DCF, causal cases, named stresses
Data/source choice	FY2025 10-K only	10-K, H1 2026 10-Q, Q2 letter, dated price, Damodaran rates	Course instruction / own analysis	sources.md; XBRL source tests
Assumption/model design	Not yet modeled	Three-statement pro-forma driving FCFF DCF	Course instruction / AI	148 model checks; known answer
Risk/limitation	Valuation burden unknown	Terminal 84% of EV; imprecise beta; 2027 growth vs bookings	Own analysis / AI review	wacc.json; named_stresses.csv

Remaining requirements are the three recordings with corrected transcripts and final access/checkpoint/receipt evidence. See Validation-and-AI-Use.pdf and submission-manifest.csv.